THE GORENSTEIN LOCUS OF A FLAT LOCALLY FINITE TYPE MORPHISM COMMUTES WITH ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E0Q (sources/stacks-duality/pinned-original-source.tex line 6683).

Allowed standard prerequisites (editorial).

- Base change of a dualizing complex over a field. Dualizing Complexes, Lemma labelled dualizing-lemma-base-change-dualizing-over-field, cited at proof lines 6710-6712. Shows the base changed complex is again dualizing.
- Flatness of the localised ring map and invertibility of the localised dualizing complex. The flat local homomorphism A_p -> (A_K)_{p_K} and the invertibility criterion used at proof lines 6714-6719. Relates Gorensteinness of the two local rings.
- Local Noetherianity of the fibres of a flat locally finite type morphism. The observation at proof lines 6690-6694. Makes the statement meaningful and stable under base change.

Source reference. Original lemma 0E0P: sources/stacks-chow/tags.txt, source lines 17544–17544. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6709 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes which is flat and locally of finite type. Then formation of the set $\{x \in X \mid f \text{ is Gorenstein at } x\}$ commutes with arbitrary base change.*

Proof. The assumption implies any fibre of f is locally of finite type over a field and hence locally Noetherian and the same is true for any base change. Thus the statement makes sense. Looking at fibres we reduce to the following problem: let

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X be a scheme locally of finite type over a field k, let K/k be a field extension, and let $x_K \in X_K$ be a point with image $x \in X$. Problem: show that $\mathcal{O}_{X_K,x_K}$ is Gorenstein if and only if $\mathcal{O}_{X,x}$ is Gorenstein.

The problem can be solved using a bit of algebra as follows. Choose an affine open $\mathrm{Spec}(A) \subset X$ containing x. Say x corresponds to $\mathfrak{p} \subset A$. With $A_K = A \otimes_k K$ we see that $\mathrm{Spec}(A_K) \subset X_K$ contains x_K. Say x_K corresponds to $\mathfrak{p}_K \subset A_K$. Let $\omega_A^\bullet$ be a dualizing complex for A. By Dualizing Complexes, Lemma 0E0P $\omega_A^\bullet \otimes_A A_K$ is a dualizing complex for A_K. Now we are done because $A_\mathfrak{p} \to (A_K)_{\mathfrak{p}_K}$ is a flat local homomorphism of Noetherian rings and hence $(\omega_A^\bullet)_\mathfrak{p}$ is an invertible object of $D(A_\mathfrak{p})$ if and only if $(\omega_A^\bullet)_\mathfrak{p} \otimes_{A_\mathfrak{p}} (A_K)_{\mathfrak{p}_K}$ is an invertible object of $D((A_K)_{\mathfrak{p}_K})$. Some details omitted; hint: look at cohomology modules. $\square$